Program : **Diploma in Communication and Computer Networking**	
Course Code : **4323**	Course Title: **Database Management Systems**
Semester : **4**	Credits: **4**
Course Category: **Programme core course**	
Periods per week: **4 (L:3 T:1 P:0)**	Periods per semester: **60**

Course Objectives:

- Understand the general concepts in database management systems.
- Understand the different types of database models.
- Understand the basics of structured query language.
- Understand the Emerging technologies in DBMS

Course Prerequisites:

Topic	Course code	Course name	Semester
Exposure to a High Level Language like C		Programming with C	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Descripti on	Duration (Hours)	Cognitive Level
CO1	To describe about Database systems	15	Understanding
CO2	To illustrate Database Design**.**	15	Applying
CO3	To write queries in SQL	15	Applying
CO4	To relate Emerging Technologies	15	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	1				
CO2	3	1	2				
CO3	3	2	1				
CO4	3	2	1				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**To describe about Database systems**		
M1.01	To learn the basics of DBMS	4	Understand
M1.02	Explain the different models in DBMS	4	Understand
M1.03	Learn about DBMS languages	3	Apply
M1.04	To learn the different DBMS component modules	4	Apply
Contents: **Database Systems** Database Systems – Data – Information – Record – Field – Need of database system - Advantages and Disadvantages - application areas – people who interact with database - Three schema Architecture - Data independence - Data models – Database Schema versus database instance – Component modules of DBMS– Classification of DBMS.			
CO2	**To illustrate database design**		
M2.01	To understand the basic relational models	7	Understanding
M2.02	To understand the relational algebra operations	8	Applying
	Series Test –I	1	

Contents: **Database Models** Relational Model Concepts – Domain – Attribute – tuple – instance – relation – relational schema – Keys – E R Model - Relational Algebra operations - select, project, Union, Set Difference, Cartesian Product and Rename – additional operations - Natural-Join, Outer Join			
CO3	**To write queries in SQL**		
M3.01	To learn the basic concepts in SQL.	4	Understanding
M3.02	To learn the commands and their usage	5	Understanding
M3.03	To learn about functions in SQL	3	Applying
M3.04	To understand database connectivity	3	Applying
Contents: **Structured Query Language** SQL – Features of SQL – Data types in SQL - CREATE TABLE command, Constraints – NULL, DEFAULT,CHECK, PRIMARY KEY, UNIQUE, referential Integrity – INSERT, UPDATE and DELETE command - SELECT statements with WHERE, ORDER BY clause - Aggregate and scalar functions in SELECT statements. Database connectivity using JDBC/ ODBC			
CO4	**To relate emerging technologies**		
M 4.01	To learn about normalization and decomposition of database	8	Applying
M 4.02	To learn about data mining and warehousing technologies	7	Understanding
	Series Test – II	1	
Contents: **Database Design and Emerging Technologies** Normalisation -Functional Dependency - Decomposition —.Introduction to Data Mining and Data warehousing (Only theoretical concepts).			

Text / Reference

T/R	Book Title/Author
T1	Database Systems – Elmasri, Navathe (Pearson) Sixth Edition.
R1	Introduction to Database Systems – ITL Education Solutions Ltd – PEARSON-2010
R2	Database system concepts - Silberschatz, Korth, and Sudarshan (TMH)-Sixth Edition
R3	SQL for professional - Swapne & Rajesh Naik
R4	Adam Fowler, NoSQL for Dummies, John Wiley & Sons, 2015
R5	NoSQL Data Models: Trends and Challenges (Computer Engineering: Databases and Big Data), Wiley, 2018

Online Resources

Sl.No	Website Link
1	https://www.w3resource.com/redis/
2	https://www.w3schools.in/category/mongodb/
3	https://www.tutorialspoint.com/cassandra/cassandra_introduction.htm
4	https://www.tutorialspoint.com/arangodb/index.htm